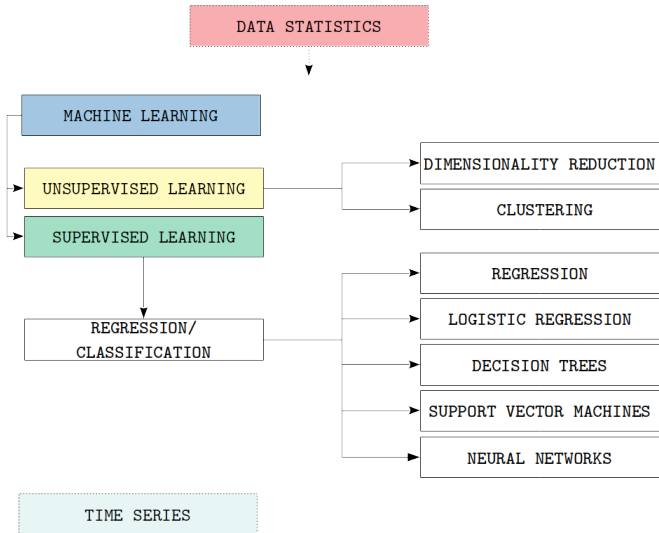


Overview

- 1 Introduction
- 2 Basic statistics
- 3 Multivariate statistics and Principal Component Analysis
- 4 Unsupervised learning: Clustering**
- 5 Supervised learning
- 6 Overview on Time Series



unsupervised vs supervised learning

- unsupervised learning

$$X = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \dots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix}$$

task: to discover interesting things about data **without knowing the true answer**

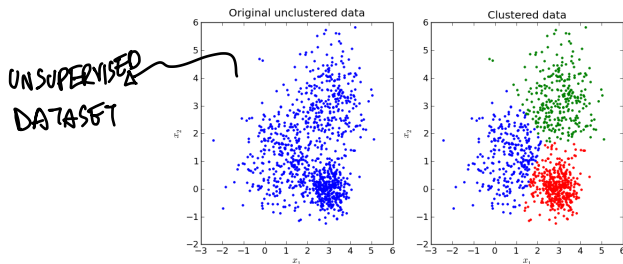
- supervised learning

$$X = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \dots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix}, Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}$$

↓ TARGET ARRAY

task: prediction of the targets

clustering



- find **clusters** (subgroups) of observations having **similar** features
- clusters $C_i, i = 1, \dots, K$ are a partition of the observations:

$$C_1 \cup C_2 \cup \dots \cup C_K = \{1, \dots, m\}, \quad C_i \cap C_j = \emptyset, \forall i \neq j$$

↗ EMPTY SET

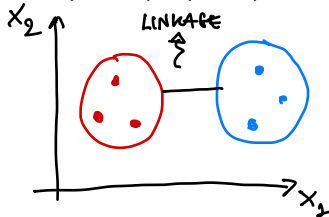
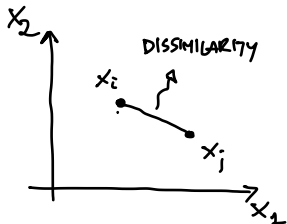
- when observations are similar (dissimilar)? domain-specific
- PCA and clustering simplify data via smaller summaries
 - ▶ PCA via low-dimension features representations
 - ▶ clustering via homogenous subgroups among observations

clustering **methods**

- **hierarchical** clustering:
the number of clusters is iteratively modified
- **K-means** clustering:
the number of clusters is decided a priori
- fuzzy clustering:
observations are assigned to all clusters with a certain probability (not to a single one)

hierarchical **agglomerative** clustering

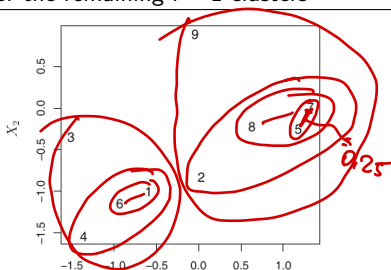
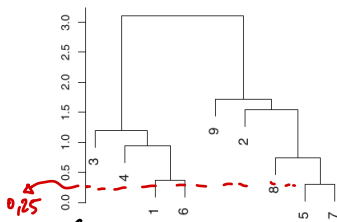
- idea: iteratively aggregate m initial clusters (one per observations) to a final single cluster (including all observations)
→ WE DON'T HAVE CORRECT ANSWERS (UNSUPERVISED) ⇒ DENDROGRAM IS HELPFUL TO ASSESS QUALITY
- dendrogram**: attractive tree-based representation of observations and clusters
- the number of clusters is decided ex post *→ DENDROGRAM CAN HELP IN DETERMINING THE # OF CLUSTERS*
- requires
 - a point to point distance (**dissimilarity**) $d(x_i, x_j)$ with $x_i, x_j \in \mathbb{R}^n$
 - a cluster to cluster distance (**linkage**) $D(C_s, C_t)$ with $s, t \in \{1, \dots, K\}$



$x_i \in \text{ROW } i$
 $\equiv \text{OBSERVATION } i$

agglomerative clustering scheme and dendrogram

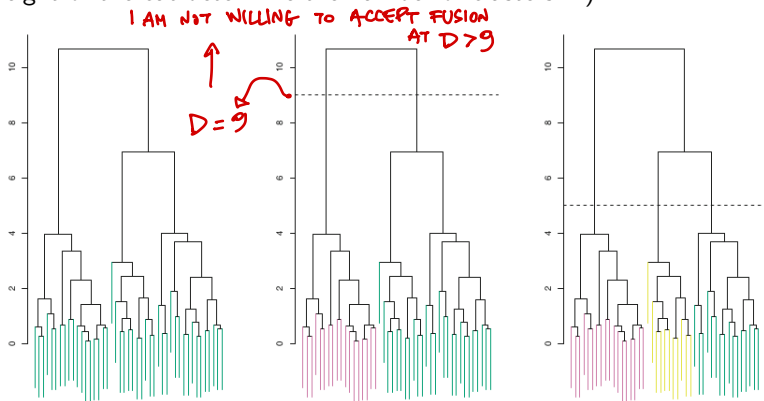
- 0 select dissimilarity d and linkage D
- 1 set the m initial clusters as the m observations and compute the $\binom{m}{2} = m(m-1)/2$ pairwise dissimilarities
- 2 for $i = m, m-1, \dots, 2$
 - a identify the pair of closest clusters according to D (or d if single observations) and fuse them
 - b compute the pairwise linkage D for the remaining $i-1$ clusters



HORIZONTAL AXIS: THE DATA POINTS
VERTICAL AXIS: THE DISTANCE (d or D) AT WHICH A FUSION IS MADE

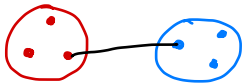
how to select the final clustering?

- make horizontal cut on the dendrogram
- subgroups below the cut are the clusters
- number of clusters = number of vertical lines crossed by the cut (height of the cut determine the number of clusters K)



linkage choices

- SINGLE LINK: $D(C_s, C_t) = \min_{x_i \in C_s, x_j \in C_t} d(x_i, x_j)$



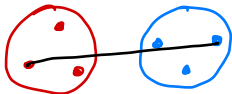
INDUCES
⇒

DISSIMILARITY $\equiv$ POINT-TO-POINT DISTANCE

ELONGATED CLUSTERS



- COMPLETE LINK: $D(C_s, C_t) = \max_{x_i \in C_s, x_j \in C_t} d(x_i, x_j)$

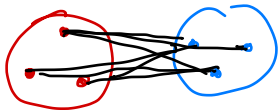


INDUCES
⇒

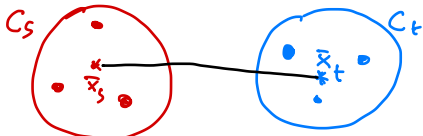
SPHERICAL CLUSTERS



- AVERAGE LINK: $D(C_s, C_t) = \frac{1}{|C_s||C_t|} \sum_{x_i \in C_s, x_j \in C_t} d(x_i, x_j)$



- CENTROID LINK: $D(C_s, C_t) = d(\bar{x}_s, \bar{x}_t)$, $\bar{x}_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$



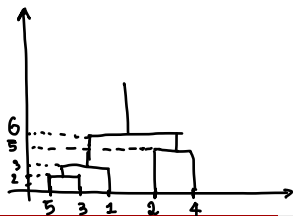
example: apply the agglomerative clustering with single link to the 5 points characterized by this pairwise Euclidean distances matrix.

	1	2	3	4	5
1	0				
2	9	0			
3	3	7	0		
4	6	5	9	0	
5	11	10	2	8	0

	1	2	{3,5}	4
1	0			
2	9	0		
{3,5}	3	7	0	
4	6	5	8	0

	{1,3,5}	2	4
{1,3,5}	0		
2	7	0	
4	6	5	0

	{1,3,5}	{2,4}
{1,3,5}	0	
{2,4}	6	0



(DISIM)

EXACT VALUES OF THE W
FEATURES NOT KNOWN

→ WE ONLY KNOWS MUTUAL
DISTANCES !!!



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some considerations on agglomerative clustering

good aspects:

- provides a nested sequence of clusterings (more informative)

bad aspects:

- sensitive to anomalous points/outliers
- irreversible: "bad" early fusions affect the whole structure of nested clusters
- arbitrary fusion choices of equally distant clusters affect the rest of the sequence
- time consuming for large datasets

K-means clustering

- the number of clusters K is given

MUTUAL DISTANCE
BETWEEN POINTS OF
↑ THE SAME
CLUSTER

- idea: a "good" clustering is one for which the **within-cluster variation** is as small as possible

↑ BARICENTER OF C_k

- each cluster C_k is characterized by its **centroid** $\bar{x}_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$



- tends to minimize the overall distance between each point and its clusters' centroid:

$$\min_{\bar{x}_1, \bar{x}_2, \dots, \bar{x}_K} W(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_K) = \sum_{k=1, K} \sum_{x_j \in C_k} d(x_j, \bar{x}_k) \rightarrow \text{NP-HARD}$$

- the algorithm is an alternation of two main steps
 - assign the points to the cluster with the closest centroid
 - determine the centroids of each cluster

HEURISTIC

↓
GOOD QUALITY
SOLUTIONS (NOT OPTIMAL)

- the algorithm stops when the two steps do not change the assignment (or do not improve $W(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_K)$) anymore

